

DSC 148 – Intro to DM: Course Project Choices

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Overview

- ❑ Team: 1 to 4 people
 - ❑ We will have a really high expectation to teams of 4 people → Nearly a top conference submission quality
- ❑ Report
 - ❑ At least 4 pages
 - ❑ Double-column, 11 pt
 - ❑ Roughly 2.5-3 thousand words + figures, tables, and equations
- ❑ Code
 - ❑ A GitHub repo
 - ❑ Working demo (bonus points up to 5%)
 - ❑ E.g., simple UI + your model: TAs or other students can give their own inputs and check the prediction

Overleaf: <https://www.overleaf.com/latex/templates/acm-conference-proceedings-master-template/pnrfvrrdbfwt>

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Five Components

- ❑ Dataset (5%)
- ❑ Predictive Task (5%)
- ❑ Model (5%)
- ❑ Literature (5%)
- ❑ Results (5%)

Dataset

- ❑ Identify a dataset
- ❑ Perform an exploratory data analysis
 - ❑ Basic Statistics
 - ❑ Properties
 - ❑ Interesting findings
- ❑ All these should motivate your model design/choice
- ❑ The dataset should be large enough (e.g., 50,000 instances in total)

Dataset – Example



- ❑ Heroes of the Storms (HOTS)
 - ❑ A 5v5 online video game
- ❑ It has massive log data available!
 - ❑ <https://www.hotslogs.com/info/API>
 - ❑ API & 30-day logs download
- ❑ EDA
 - ❑ Any frequent combinations of heroes?
 - ❑ Which hero is the “weakest”?
 - ❑ ...
 - ❑ Some analysis: <https://github.com/shangjingbo1226/HOTS-Analysis/blob/master/combo-analysis.ipynb>



Predictive Task

- ❑ Identify a predictive task based on your dataset
- ❑ How will you evaluate different models in this task?
- ❑ What are the baseline models you want to compare with?
 - ❑ Why do you think they are appropriate?
 - ❑ Why do you think your model can outperform them? Or say, what are their drawbacks?

Predictive Task – Example



- ❑ Heroes of the Storms (HOTS)
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- ❑ It has massive log data available!
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 - ❑ API & 30-day logs download
- ❑ Given the hero picks of two teams (of similar levels), which team will win?
 - ❑ A classification problem!

Predictive Task – Example



- ❑ Given the hero picks of two teams (of similar levels), which team will win?
 - ❑ A classification problem!
- ❑ Evaluation
 - ❑ Accuracy, F1
- ❑ Baselines
 - ❑ Logistic regression: Assume it's a linear combination of selected heroes
 - ❑ Naïve Bayes: Assume the selected heroes are conditionally mutual independent to each other

Model

- ❑ What is the model that you propose to attack this task?
 - ❑ It's fine to use models that were described in class here
 - ❑ Explain and justify your choice/proposal What are the features you designed for your model?
 - ❑ Any unsuccessful tries?
- ❑ How will you optimize your model?
 - ❑ It's fine here to call any 3rd-party libs
- ❑ Did you encounter any troubles?
 - ❑ Scalability? Overfitting?

Model – Example



- ❑ How can we represent the input?
 - ❑ One hot encoding?
 - ❑ Heroes have no order?
 - ❑ Meta-data (e.g., hero types, user ratings, ...)?
 - ❑ Battlegrounds (Maps)?
 - ❑ ...
- ❑ Application?
 - ❑ Real-time demo
 - ❑ Retrieve all users' info (rating, history, ...)
 - ❑ Suggest some picks
 - ❑ How to update when there is a new hero or a rework?

Literature

- ☐ Has your dataset/task been studied by others before?
- ☐ How the dataset was used?
- ☐ Are you working on a brand-new task?
- ☐ How are other people attacking the same/similar tasks?
- ☐ What is state-of-the-art method in this task or related tasks?
- ☐ Are your conclusions similar or different from existing work?
- ☐ What's the major novelty of your work?
- ☐ ...

Literature – Example



- There are some apps doing this task already
 - Looking into that and check their performance, if possible
- There might be similar tasks (e.g., DoTA AI?)
 - What are their solutions?
- ...

Results

- ❑ Does your proposed method outperform the baselines?
 - ❑ Why your model can outperform?
 - ❑ Or why your model fails?
- ❑ Whether the gap is significant?
- ❑ Are all features you designed effective?
- ❑ How shall one set the hyper-parameters of your model?
- ❑ What are the major takeaways (i.e., conclusions)?
- ❑ ...

Results – Example



- ❑ Performance comparison different methods
 - ❑ Baselines + Your proposed model
- ❑ Ablation study
 - ❑ What if some of the features/designs of your proposed model degenerates?
- ❑ Case Study
 - ❑ Some interesting cases when your model performs very well/poor
- ❑ Parameter Sensitivity
 - ❑ How do you decide hyperparameters?
 - ❑ Is the result sensitive to these hyperparameters?
- ❑ ...